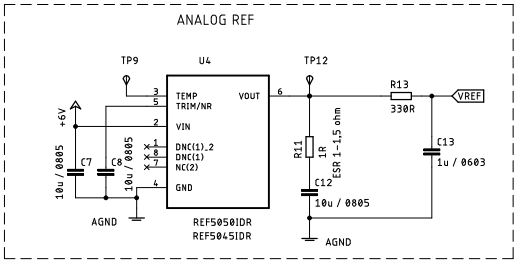
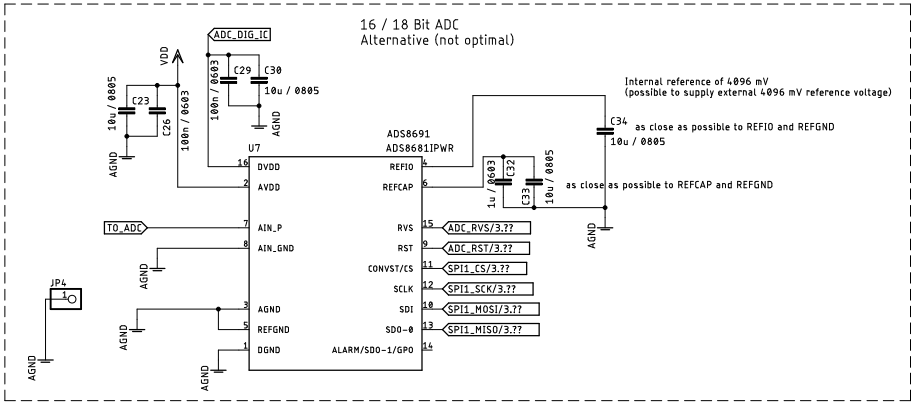
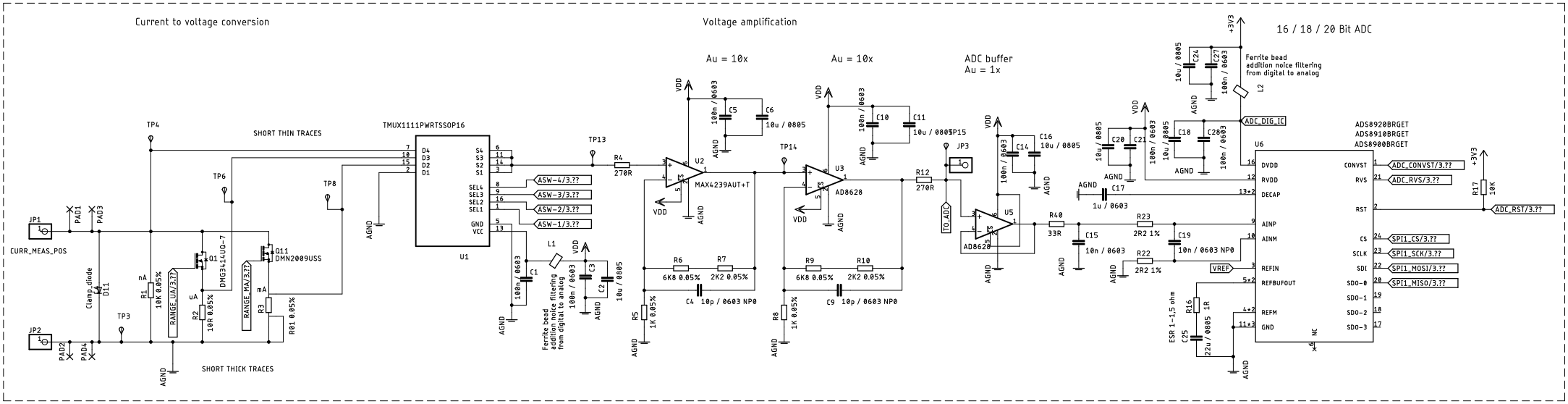

ANALOG REF
TP9
U4
TP12
TEMP
TRIM/NR
VIN
DNC(1),2
DNC(1)
NC(2)
GND
VOUT
R13
330R
VREF
+6V
C7
C8
10u / 0805
AGND
REF5050IDR
REF5045IDR
R11
1R
ESR 1-1,5 ohm
C12
10u / 0805
C13
1u / 0603


16 / 18 Bit ADC
Alternative (not optimal)
ADC_DIG_IC
VDD
C29
C30
10u / 0805
100n / 0603
C23
C26
10u / 0805
100n / 0603
AGND
U7
ADS8691
ADS8681IPWR
DVDD
AVDD
AIN_P
AIN_GND
AGND
REFGND
DGND
REFIO
REFCAP
RVS
RST
CONVST/CS
SCLK
SDI
SDO-0
ALARM/SDO-1/GPO
TO_ADC
JP4
Internal reference of 4096 mV
(possible to supply external 4096 mV reference voltage)
C34
10u / 0805
as close as possible to REFIO and REFGND
C32
C33
1u / 0603
10u / 0805
as close as possible to REFCAP and REFGND
ADC_RVS/3.??
ADC_RST/3.??
SPI1_CS/3.??
SPI1_SCK/3.??
SPI1_MOSI/3.??
SPI1_MISO/3.??


Current to voltage conversion
Voltage amplification
16 / 18 / 20 Bit ADC
JP1
JP2
PAD1
PAD3
PAD2
PAD4
CURR_MEAS_POS
Clamp_diode
D11
TP4
TP3
TP6
TP8
SHORT THIN TRACES
SHORT THICK TRACES
nA
R1
10K 0.05%
RANGE_nA/3.??
uA
R2
10R 0.05%
Q1
DMG3414UQ-7
mA
RANGE_mA/3.??
R3
R01 0.05%
Q11
DMN2009USS
AGND
TMUX1111PWRTSSOP16
U1
D4
D3
D2
D1
S4
S3
S2
S1
SEL4
SEL3
SEL2
SEL1
GND
VCC
ASW-4/3.??
ASW-3/3.??
ASW-2/3.??
ASW-1/3.??
TP13
R4
270R
C1
100n / 0603
L1
Ferrite bead
addition noice filtering
from digital to analog
C3
100n / 0603
C2
10u / 0805
VDD
Au = 10x
U2
MAX4239AUT+T
C5
100n / 0603
C6
10u / 0805
R6
6K8 0.05%
R7
2K2 0.05%
C4 10p / 0603 NP0
R5
1K 0.05%
TP14
Au = 10x
U3
AD8628
C10
100n / 0603
C11
10u / 0805
R9
6K8 0.05%
R10
2K2 0.05%
C9 10p / 0603 NP0
R8
1K 0.05%
TP15
JP3
R12
270R
TO_ADC
ADC buffer
Au = 1x
U5
AD8628
C16
10u / 0805
C14
100n / 0603
R40
33R
C15
10n / 0603
R23
2R2 1%
C19
10n / 0603 NP0
R22
2R2 1%
C17
1u / 0603
+3V3
C24
10u / 0805
C27
100n / 0603
Ferrite bead
addition noice filtering
from digital to analog
L2
C20
10u / 0805
C21
100n / 0603
C18
10u / 0805
C28
100n / 0603
ADC_DIG_IC
U6
ADS8920BRGET
ADS8910BRGET
ADS8900BRGET
DVDD
RVDD
DECAP
AINP
AINM
REFIN
REFBUFOUT
REFM
GND
CONVST
RVS
RST
CS
SCLK
SDI
SDO-0
SDO-1
SDO-2
SDO-3
NC
ADC_CONVST/3.??
ADC_RVS/3.??
ADC_RST/3.??
SPI1_CS/3.??
SPI1_SCK/3.??
SPI1_MOSI/3.??
SPI1_MISO/3.??
R17
10K
R16
1R
ESR 1-1,5 ohm
C25
22u / 0805
VREF